

Question ID: f1c81b3b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The exponential function g is defined by $g(x) = 19 \cdot a^x$, where a is a positive constant. If $g(3) = 2,375$, what is the value of $g(4)$?

Correct Answer: 11875

Rationale

The correct answer is **11,875**. It's given that the exponential function g is defined by $g(x) = 19 \cdot a^x$, where a is a positive constant, and $g(3) = 2,375$. It follows that when $x = 3$, $g(x) = 2,375$. Substituting **3** for x and **2,375** for $g(x)$ in the given equation yields $2,375 = 19 \cdot a^3$. Dividing each side of this equation by **19** yields $125 = a^3$. Taking the cube root of both sides of this equation gives $a = 5$. Substituting **4** for x and **5** for a in the equation $g(x) = 19 \cdot a^x$ yields $g(4) = 19 \cdot 5^4$, or $g(4) = 11,875$. Therefore, the value of $g(4)$ is **11,875**.